Diffusion models for cosmology : Component separation and cosmological inference from mixtures of dust and CMB

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ÉCOLE POLYTECHNIQUE

Contributions

- A generative diffusion model of simulated interstellar dust emission maps:
  - Samples agree visually and quantitatively with the training set → histograms, power spectrum and Minkowski functionals are preserved.
- Diffusion-based source separation from an additive mixture of dust emission and CMB:
  - Dust and CMB maps are statistically reconstructed → power spectra match overall.
  - Power spectrum discrepancies at small scales for dust.
- Diffusion-based inference of cosmological parameters from mixture of dust and CMB.

Background

- Cosmic Microwave Background (CMB) data contaminated by the thermal emission of interstellar dust → need for a realistic model of dust.
- Diffusion models have proved extremely powerful for a variety of natural images generative tasks (e.g., prompted generation, inpainting, deblurring, linear inverse problem).

Method: Diffusion Model with non-white Gaussian noise

Our Problem

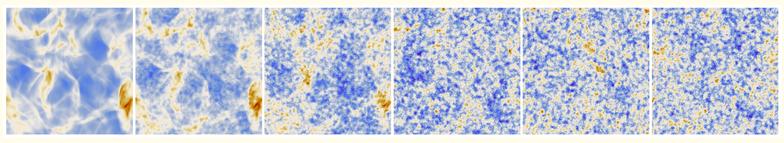
- We observe $\mathbf{y}_{\text{obs}} = \mathbf{x}_{\text{dust}} + \varepsilon_{\text{CMB}}$ with $\varepsilon_{\text{CMB}} \sim \mathcal{N}(0, \sigma^2 \boldsymbol{\Sigma}(\phi))$ and $\phi \sim p_\phi$

  with ϕ the cosmological parameters, p_ϕ a prior distribution and $\boldsymbol{\Sigma}(\phi)$ the covariance of the CMB. σ is used to control the relative strength of the CMB with respect to dust.
- Goal: sample $p(\mathbf{x}_{\text{dust}}|\mathbf{y}_{\text{obs}}, \phi)$ (component separation) and $p(\phi|\mathbf{y}_{\text{obs}})$ (cosmological inference).

Mathematical formulation

- Diffusion models (DM) rely on two stochastic differential equations (SDE) [6]:
  - forward SDE → stochastically maps our target distribution to a Gaussian distribution.
  - reverse SDE → maps a Gaussian to our target distribution using the score function of the noisy distribution $\mathbf{s}(\mathbf{x}_t, t) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$.

$$\mathbf{X}_0 \xrightarrow{\mathrm{d}\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\,\mathrm{d}t + \boldsymbol{\Sigma}_t(\phi)^{1/2}\,\mathrm{d}\mathbf{W}_t} \mathbf{X}_1$$



$$\mathbf{X}_0 \xleftarrow{\mathrm{d}\mathbf{x}_t = \{\mathbf{f}(\mathbf{x}_t, t) - \boldsymbol{\Sigma}_t(\phi)\,\mathbf{s}(\mathbf{x}_t, t)\}\,\mathrm{d}t + \boldsymbol{\Sigma}_t(\phi)^{1/2}\mathrm{d}\mathbf{W}_t} \mathbf{X}_1$$

Introducing $\beta(t) > 0$, $\mathbf{x}_t \in \mathbb{R}^d$ with $t \in [0, 1]$, and $\boldsymbol{\Sigma}$ a covariance matrix, we define:

$$\mathbf{f}(\mathbf{x_t}, t) = \frac{-\beta(t)}{2}\mathbf{x_t}, \quad \boldsymbol{\Sigma}_t(\phi) = \beta(t)\boldsymbol{\Sigma}(\phi).$$

- A score network $\mathbf{s}_\theta(\mathbf{x}, t)$ learns $\mathbf{s}(\mathbf{x}_t, t)$, using the denoising score matching loss [7].

Component separation and cosmological inference, leveraging a Gaussian CMB

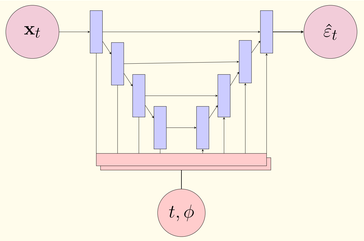
- Sampling $p(\mathbf{x}_{\text{dust}}|\phi, \mathbf{y}_{\text{obs}}) \iff$ sampling $p(\mathbf{x}_0|\mathbf{x}_{t^*})$ for some adequate time t^*.
- Gibbs sampling for $p(\phi|\mathbf{y}_{\text{obs}})$: we create a Markov chain by alternatively sampling $p(\mathbf{x}_{\text{dust}}|\phi, \mathbf{y}_{\text{obs}})$ with the diffusion model and $p(\phi|\mathbf{x}_{\text{dust}}, \mathbf{y}_{\text{obs}})$ with a Hamiltonian Monte Carlo sampler.

Data

- Dust simulations from CATS database [1]→ we build total intensity maps.
- CMB power spectrum computations accelerated by an emulator trained on CAMB realizations.
- Work restricted to 2 cosmological parameters: $\phi = (H_0, \omega_b)$. Remaining parameters are fixed.

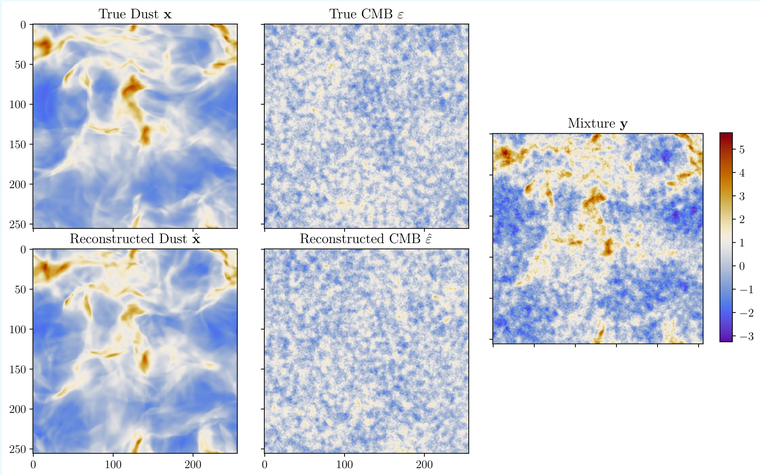
Model details

- 4 level UNet with ResBlocks [3].
- Noise cutoff $\beta(0) \geq \beta_{\min}$ and time cutoff $t_{\min} > 0$ needed for stability ⇒ can impact higher frequencies.
- $\boldsymbol{\Sigma}$ has a high condition number → learning $\mathbf{m}_\theta \propto \boldsymbol{\Sigma}(\phi)\,\mathbf{s}_\theta$ → stabilizes training.



Result 1: A component separation preserving summary statistics

We first explore the sampling of $p(\mathbf{x}_{\text{dust}}|\mathbf{y}_{\text{obs}}, \phi)$:



- Dust is well reconstructed. Discrepancies of the reconstruction can be found at small scales.
- Variance of the CMB reconstruction process lies at large scales → poor appearance.

Preservation of summary statistics

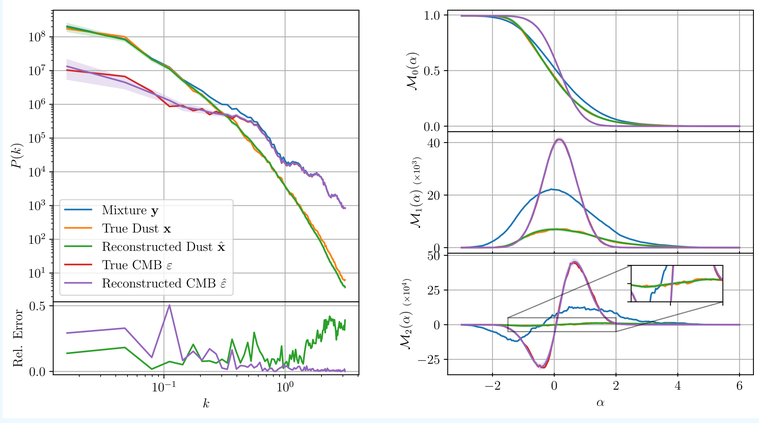


Figure 1. Summary statistics of the reconstructed signals: power spectrum (left) and Minkowski functionals (right).

Summary statistics match overall:

- Summary statistics of the reconstructions agree overall with those of the true maps.
- Slight discrepancies at small scales for reconstructed dust, other scales match well.
- Larger variance at large scales for the CMB.

Result 2: Cosmological Inference (preliminary results)

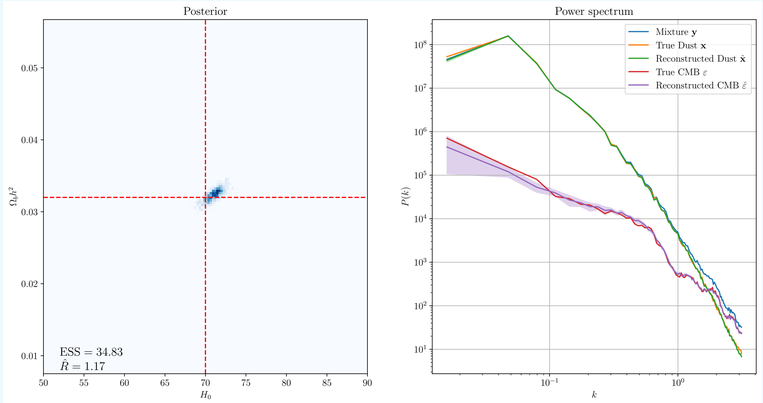


Figure 2. Cosmological inference with $\sigma = 0.1$ (Weak CMB)

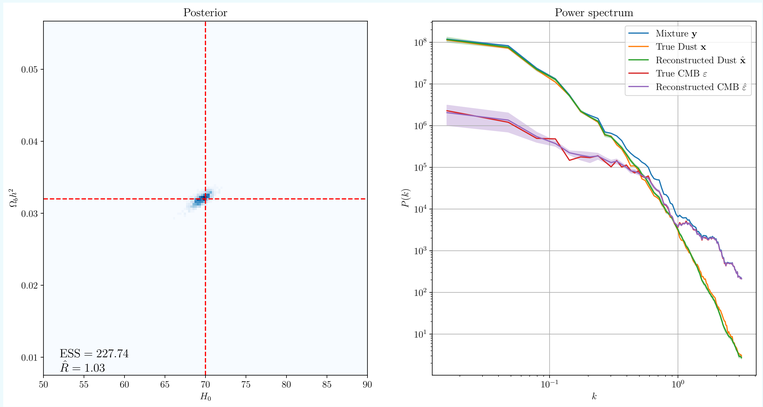


Figure 3. Cosmological inference with $\sigma = 0.3$ (Strong CMB)

- First results are encouraging, Gibbs sampling converges quickly.
- Statistics of the reconstruction agree with the true statistics.
- *Work in progress* : validate our inference method, thorough analysis for different levels of CMB.

Perspectives

- Application to other Gaussian denoising and inference problems with natural images.
- Extension to polarization/multi-frequency data, extension to the sphere.
- Extension to non-Gaussian CMB, taking into account gravitational lensing.

References

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